

Intrusion detection models for IOT networks via deep learning approaches

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ABSTRACT

The Internet of things (IoT) has gained more attention in recent years because of its ubiquitous operations, connectivity, methods of communication, and intelligent decisions to evoke activities from various devices. Therefore, artificial intelligence techniques have been integrated into all aspects of the Internet of Things and making life more comfortable in various ways. A novel deep learning model named Device-based Intrusion Detection System (DIDS) was proposed in the second phase. This DIDS learning model incorporates the prediction of unknown attacks to handle the computational overhead in large networks and increase the throughput with a low false alarm rate. Our proposed algorithm has been evaluated with standard algorithms, and the results show that it detects attacks earlier than standard algorithms. The computational time has also been reduced, and 99% of accuracy has been achieved in detecting the attacks.

1. Introduction

IoT networks are becoming more vulnerable to security attacks as their popularity rises. The threat of cyber-attacks is rapidly becoming one of the most severe threats to IoT security. These attacks occur in many ways and typically target anomalous assets, affecting one or more IoT devices that can be used as “resources” or “platforms” for the attack, such as distributed denial-of-service, Man middle attack, spoofing, and data leakage. As a result, securing IoT devices and designing intrusion-resistant IoT networks become increasingly important to safeguard data.

In this booming digital era, each device is interconnected with the physical world through the internet of things (IoT) [1]. Significant technologies such as fog, cloud computing, artificial intelligence, and 5G progress in IoT have become the backbone of this digital world. These progressions have driven the advancement of smart cities, intelligent vehicles, intelligent factories, etc., that target advancing people's lifestyles productively and cost-effectively. In addition to this, IoT nowadays is called as “Internet of Everything” (IoE) because it deals with all categories of smart devices. As per the global market survey, in 2025 [2], the total count of connected IoT devices may reach 21.5 billion. Moreover, IoT systems generally store and handle data dispersed instead of the highly centralized method of aggregating storage and processing resources in massive data clusters [3].

In the present situation, tremendous advancements in computer technology have fundamentally altered our everyday lives and rendered us dependent on them. Many of their operations are carried out via network services, whether they are small businesses, major corporations, private citizens, or government organizations. Our computer systems are subject to an increasing number of risks since Internet use has grown so rapidly. Any weakness in the computer platforms and network devices might expose the network system to different assaults and can have disastrous results. However, protecting organizations' and corporations' networks from attacks and keeping their secrets and important consumer information from leaking out is a nightmare for security administrators [4]. Organizations should focus on innovative techniques to assist human analysts when dealing with surveillance, prevention, detection, and response to cyber security incidents and potential attacks/intrusions on their networks. The

the problem of how to effectively detect and prevent intrusions in time is very challenging. Conventional network intrusion detection systems (NIDS) are signature or rule-based approaches that have not been adequate for the fastgrowing network and unable to deal with attacks of their growing volume, complexity, and deflation. The process of monitoring a computer system or network's activity and analyzing it for indications of potential events, such as violations or suspicious behaviour of computer security regulations, permissible usage policies, or

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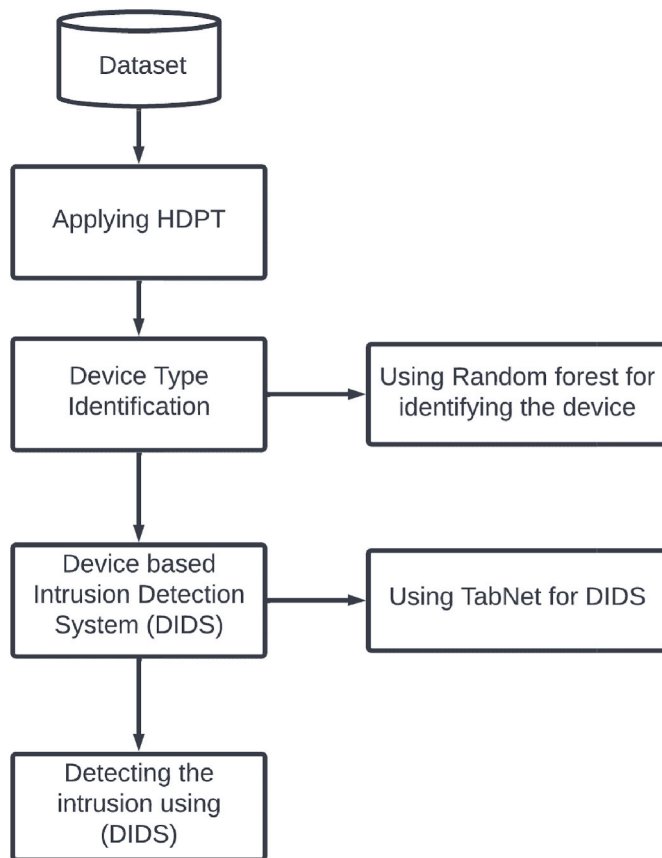


Fig. 1. Over all approach for DIDS.

standard security practises, is known as intrusion detection. There are too many potential causes of events, including harmful software (such as worms and spyware), online hackers who get remote access to systems, and privileged individuals who abuse their rights or try to maximise powers for which they are not authorized. Software that maintains the intrusion detection method is known as an intrusion detection system (IDS). The primary focus of the IDS is the identification of possible events. IDS would be able to tell whether a system has been successfully infiltrated by a hacker using system vulnerabilities [5].

Many IDS may be configured with a set of firewall rules, such as settings, to help them identify network traffic that violates the organization's security or acceptable usage requirements. Some IDSs can monitor data transfer and spot unusual ones, such as copying a large database to a user's computer. IDSs are primarily focused on identifying potential occurrences. Security administrators may get reports from the IDS and immediately begin incident response procedures to lessen the impact of the occurrence. The IDS also maintains log data that system administrators may utilize. Multiple intrusions Systems for detection are set up to spot security policy violations. IDS is now a crucial part of computer security since it may identify assaults before they have a chance to do significant harm, and intrusion detection technology has received a lot of attention recently [6,7].

1.1. Methods for detecting intrusions

IDS technologies detect intrusions using a variety of approaches. The two main types of detection approaches are anomaly-based detection and intrusion detection based on signatures. To properly identify intrusions, the majority of IDS solutions use a variety of detection algorithms, either alone or in combination.

1.1.1. Detection based on signatures

The database of signatures from known assaults is kept through signature-based detection, which is also known as misuse-based detection. An identifiable threat has a pattern known as a signature. When data is collected from the audit unit, it is compared to the database, and alarms may be triggered if a match is found. These patterns represent suspicious groups of operations or sequences that may be hazardous and are kept in databases. The main advantage of this system is that it quickly learns patterns and fingerprints and, if acquainted, understands network behaviour. Maintaining assaults whose patterns have already been recorded in the database is more efficient. These approaches' main drawback is that they can only forecast intrusions whose attack patterns have already been recorded in the database. A unique signature must be created for each assault. The lack of attack signatures in their database makes it clear that these sorts of IDSs are unable to identify zero-day assaults. Furthermore, the signature is only effective against attacks with set behavioural patterns; it is ineffective against assaults involving humans or attacks with attacks that naturally exhibit self-modifying behaviour [8].

When the client uses cutting-edge technological platforms like no operation (NoP) generators or encoding and decoding payloads, these detection systems are likewise ineffective. The effectiveness of signature-based systems declines when dynamic structures are developed for various variants. The engine's performance can decline as the number of signatures rises. But the nicest feature of this IDS approach is how seldom false alarms occur [9].

1.1.2. Detection of anomalies

The second kind of approach is anomaly-based IDS, which is also sometimes referred to as behaviour-based IDS. Instead of storing the signatures of known assaults, these systems learn the typical behaviour of individuals, hosts, network connections, or applications, i.e., store the signature of the typical behaviour. By tracking the features of usual activity over time, behaviour profiles are created. Profiles may be created for a variety of behavioural characteristics, including the quantity of emails sent by a user, the number of unsuccessful tries to log in to a host, and the amount of processor consumption during a certain time period. As it creates the first profiles for its detection algorithms, an intrusion detection system may also spot malicious behaviour. Any departure from the usual is considered suspicious, and an alert is sent off. These systems function on the presumption that any anomalous behaviour or activity diverges greatly from the norm [10].

These systems are, by nature, effective in identifying zero-day assaults.

Profiles might be static or dynamic for anomaly-based detection. Unless the system is expressly instructed to build a new profile, the static profile that has already been produced remains unmodified. As new instances are noticed, a dynamic profile is regularly updated. As systems and networks evolve over time, the corresponding measurements of normal behaviour often change as well. A static profile must be routinely produced since it will ultimately stop being reliable.

While dynamic profiles do not experience this issue, they are nonetheless vulnerable to evasion techniques used by attackers. An attacker could sometimes engage in small amounts of malicious activity, then gradually increase both the frequency and the volume of activity. If the change rate is too slow, the detection system can mistake the malicious activity for regular behaviour that is already part of its profile [11].

The main advantages of anomaly-based detection techniques are their high sensitivity to new assaults. Imagine if a computer is infected with a new kind of malware that can use up the system's processing power, send a lot of emails, start a lot of network connections, and engage in other behaviours that are quite different from the established profiles for the system [12].

These systems have a greater risk of false alarms since any change from routine behaviour is seen as an incursion. Researchers are also focused on these sorts of devices to reduce the false alert rate. The

successful application and validation of rule sets across all protocols is essential for the system to operate as intended. Anomaly-based detection has certain drawbacks, one of which is the need to describe the rule sets. Another noteworthy issue is that it is never simple to analyze and ascertain why a certain warning has been generated, as well as to confirm that an alert is accurate. Fig. 1 shows the methodologies and approaches for anomaly detection. The most popular IDS type and the most extensively researched subject in the literature by researchers are anomaly detection methods [13]. This kind of intrusion detection technique may also be effective even in the absence of signature pattern matches and when traffic patterns are abnormal.

1.2. Key challenges of IoT networks

The IoT helps businesses and enterprises operate productively with enhanced ease and efficiency. However, there are key challenges to be addressed in developing an IoT environment and adapting to it completely, as listed below:

1.2.1. Security

IoT devices connected to the internet are vulnerable to attack; there are a variety of attacks that threaten the security of IoT devices, making them vulnerable to attacks.

1.2.2. Awareness

IoT is capable of improving the lives of many people; despite the technology, it is receiving very little attention because only a few consumers are aware of what it means, and due to the lack of both awareness and value perception of IoT.

1.2.3. Interoperability

IoT systems face significant challenges when it comes to interoperability since all devices are heterogeneous when they are linked and integrated. In addition, these components are from different vendors, independently developed, and use diverse communication protocols.

1.2.4. Business model

The Internet of Things has a significant impact on the economy, transforming many businesses into digital businesses and enabling enterprises to automate manual processes and create new business models. However, there is still a question about how industries can monetize their services with IoT solutions.

1.2.5. Connectivity

The number of heterogeneous IoT devices connected to the network increases, the existing Internet infrastructure collapses, and connection problems occur. Communication plays a vital role in sending data from sensors, storing them in IoT server modules, and performing automation to provide feedback to actuators or applications.

1.2.6. Big data

Massive amounts of data are generated continuously and simultaneously by many devices, referred to as Big Data. IoT deals with large volumes and different types of data, which will cause significant challenges in terms of time, resources, processing abilities.

1.2.7. IoT threats categorization

IoT threats [4] in intrusion detection based system are broadly classified as.

- i. IoT threats based on layers
- ii. IoT

1 IoT threats based on layers

The standard IoT architecture is visualized in different layers namely

perception layer, network layer and application layer as shown in Fig. 1.

i) The perception layer

This layer is primarily concerned with object identification and collecting data from RFID tags, QR codes, cameras, terminals, wireless sensor networks, 2D barcode labels, and other types of sensors. The following are possible threats to the perception layer:

1.2.8. Authentication

The authentication of the sensor network is threatened by outside attacks such as eavesdropping, replay attacks, spoofing, and packet amendment.

1.2.9. Network availability attacks

A denial of service (DoS) attack impacts the sensor community. Sensor networks are divided into layers so that any sensor layer could be at risk of this type of DDoS attack.

1.2.10. Service integrity

An attacker uses this method to make the network accept incorrect data values.

1.2.11. Jamming

Signal jamming or radio jamming is an attack that interferes with communication between nodes.

1.2.12. Eavesdropping

In this technique, the attacker secretly monitors the conversation between the sender and the receiver to obtain the information.

1.2.13. Replay attack

Attackers interrupt communication between tags and readers, and duplicate titles are used to match authentication sequences.

Man-In-The-Middle (MITM) attack.

It occurs when an intruder enters into communication between a user and an application, either to eavesdrop or spoof one of the parties, creating the appearance of normal information flow will occur.

ii) The network layer

Network layers work similar to neural networks and are sometimes called the brain of IoT. In general, this layer sends and processes information from the perception layer. The system includes an Internet network, an information distribution hub, an intelligent data processing centre, and a network management centre. This layer's primary job is to process and forward data received from the perception layer. 6.

1.2.14. Selective forwarding

The attacker selectively forwards some anomalous messages to reveal its misdeed.

1.2.15. Sybil attack

An attacker can launch the Sybil attack at more than one location simultaneously as a single malicious node. This type of attack involves a malicious device taking on multiple identities in a network. These attacks affect fault-tolerant systems.

1.2.16. Sinkhole attack

In this case, the network becomes overloaded, and node energy consumption increases in sensor networks, the network becomes susceptible to denial-of-service attacks.

1.2.17. Wormhole attack

This kind of Denial of Service (DoS) attack moves data bits on a low-latency link from their original position.

1.2.18. Acknowledgment

Flooding Sensor-based systems frequently require a routing mechanism to deliver exposures. Malicious nodes then use these acknowledgments to transmit false information to target neighbors.

1.2.19. The application layer

The layer is responsible for delivering application-specific services to the end user. It deals with various applications where the IoT can be deployed, such as smart agriculture, smart homes, smart cars, and intelligent healthcare.

2. Literature survey

Numerous intrusion detection systems have been created by researchers during the last ten years. Some of these were designed to interact with host-based intrusion detection systems, while others were network-based. The systems that are being shown combine HIDS/NIDS with signature- and anomaly-based techniques to form hybrid systems. Computational intelligence is a type of intelligence system that is used in intrusion detection systems. The primary objective of computational intelligence is to offer answers to challenging situations in the actual world. To build a more reliable IDS, a successful IDS must incorporate soft computing approaches in addition to normal mathematical procedures and analytical methodologies. In this paper, we examine three key areas of study that have substantial implications for the framework that is provided. We begin by reviewing current machine learning and deep learning shallow learning intrusion detection technologies. The second portion of this article examines current hybrid intrusion detection systems. Finally, we present a few conclusions and an analysis of the literature reviews [14].

Numerous research has been done on data pre-processing in IoT devices, however, most of them just provided solutions for certain scenarios without stating why they selected the particular filter criterion they did. On the one hand, it can be difficult to determine whether external influences contributed to the corruption of the data. Therefore, as stated in Ref. [15], the filtering stage must be finished before the outlier detection stage. However, filtering can get over physical restrictions by giving a clean dataset to apply many stages. The literature review revealed two final outstanding issues in data filtering: the absence of a sensor taxonomy related to the data collection procedure and the determination of appropriate sample rates for each type of sensor.

Deep learning has produced outstanding outcomes for picture categorization challenges in the deep learning, however, frequently requires that all sample data contain the label or category information pertinent to the sample, which is challenging to do in the real world. To address the issue of inconsistent distribution of source data and target data, a migration learning approach is suggested. Deep neural networks, which have strong processing capabilities and effects for tasks like image classification, can be fitted using migration learning. However, current migrating learning algorithms have a limited range of applications and mainly rely on manual experience [16].

Researchers have focused some effort on migration learning due to the difficulty of real-world problems after deep learning achieved considerable success in image, speech, video, and integrated tasks. Research findings on migration learning have recently been published in several prestigious worldwide academic conferences, including ICML, SIGKDD, NIPS, ICDM, etc. Migration learning has attracted a lot of interest. Making algorithms that pass the Turing test is the core objective of artificial intelligence. The relationship between this fundamental objective and self-learning is in dissociable, and migration learning is a prerequisite for self-learning (Pan & Yang, 2010; Wei, Zhang, & Yan, 2014). Inductive learning, direct push learning, and unsupervised learning are the three main categories that can be used to categorize migration learning [17].

2.1. Related study with deep learning

Deep learning, a subset of machine learning that has gained popularity recently, has been used for intrusion detection. The majority of research has shown that deep learning outperforms conventional techniques. This section discusses the literature review of the single- and hybrid-deep learning approaches that have been used by researchers.

Accelerated Deep Neural Network (DNN) was created by Sasanka Potluri et al. to detect anomalies in network data. This architecture is made up of procedures for pre-training and fine-tuning. There are two autoencoders used in the pre-training procedure. One autoencoder with 20 neurons chose 20 features from the NSL-KDD dataset's 41 features, while a second autoencoder with 10 neurons chose 10 features. Following the pre-training procedure, phase one of fine-tuning uses a softmax classifier to categorize the multiclass assaults. The complete produced DNN is further fine-tuned via backpropagation in step two on every concealed layer. The characteristics in the intermediate layers are improved by this guided backpropagation. The model is therefore more applicable to the problem of intrusion detection [18].

Using the KDDCup'99 dataset, Jihyun Kim et al. build an IDS model based on deep learning called the Long Short Term Memory recurrent neural network (LSTM-RNN). 10% of the data was utilized to train and test the model. The DoS attack is given more weight in the data ratio of the assaults. Even though the FAR is somewhat higher than the others, the model fared best in terms of DR and accuracy [19].

Suitable optimizer for Long Short Term Memory recurrent neural network was discovered by Thi-Thu-Huong Le et al. . Six optimizers, including RMSprop, adagrad, adadelta, adam, adamax, and nadam, are used to build the model. On the KDDCup'99 dataset, the model performance is measured in two situations. The average percentage categorization of assault kinds is calculated in Case 1. Case 2 computes parameters including accuracy, detection rate, false alarm rate, recall, and efficiency to assess categorization performance. With a learning rate of 0.002, the model outperforms in both scenarios.

Recurrent Neural Network IDS (RNN-IDS) for binary and multi classification was suggested by Chuanlong Yin et al. [4]. There are two components to the methodology: forward propagation and backward propagation. Calculating the output values is the responsibility of forward propagation, and transmitting the residuals that accumulated to update the weights is the responsibility of back propagation. On the NSL-KDD dataset, the model assessed the outcomes. Performance of the technique in comparison to classical classification models like J48 and Naïve accuracy, detection rate, and false positive rate between Bayesian and Random Forest [20].

For unsupervised feature learning, Nathan Shone et al. presented the nonsymmetric deep autoencoder (NDAE). An autoencoder with non-symmetrical multiple hidden layers is called NDAE. Fundamentally, the suggested model uses just the encoder phase rather than the symmetric encoder-decoder paradigm (non-symmetric). This is because calculation time may be cut down with little effect on accuracy and productivity. NDAE functions well as a scalable hierarchical unsupervised feature extractor for high-dimensional input. The feature learned data by the NDAE is classified using the random forest (RF). On the KDD'99 and NSL-KDD datasets, the model is assessed with high levels of accuracy, precision, and recall along with a short training period [21].

A few-shot deep learning strategy was provided by Md Moin Uddin Chowdhury et al. [7] to identify minority attack types in the KDD'99 and NSL-KDD datasets. Deep Convolutional Neural Network (CNN) is first taught to identify intrusions. The CNN architecture is supplied the pre-processed train and test data. The intermediate features are retrieved, and the SVM and k-NN are then given those features as input. For the k-NN classifier, they took into account 1 neighbour. On the KDD and NSL-KDD datasets, the model achieved cutting-edge results [22].

A unique strategy known as the Spectral Clustering and Deep Neural Network (SCDNN) method was introduced by Tao Ma et al. [8]. To begin with, the dataset is based on sample similarity utilizing the cluster

centre, or spectral cluster, split into k subsets. Next, based on similarity characteristics, the distance between data points in a test and train set is calculated. Each DNN has a five-layer design, including an input layer, two hidden layers, a softmax layer, and an output layer. In this model, the top layer is a five-dimensional output vector layer, while the two hidden layers each learn features from clustered data. The training subset data from the k th cluster centre produced by the SC algorithm during the clustering phase is fed into the k th DNN in turn. Sub-DNN1 through sub-DNN k are the designations for the trained sub-DNN models. Following the completion of the k training data subsets, the k sub-DNNs are fed the k testing data subsets. The final output of each sub-DNN is integrated, and the positive detection rates are assessed. The KDDCup'99 and NSL-KDD datasets were used to assess the model [23].

Deep learning was presented by Shahriar Mohammadi et al. [9] for identifying the best characteristics. It uses the z-score to first normalise the data. A distribution with a mean of zero and a standard deviation characterizes this approach. The second step involves the reduction of a vast scale of features using a deep learning autoencoder model. The achieved characteristics are employed in the last step of the algorithm as the final classification function, which can distinguish between normal and aberrant traffic. On the KDD and NSL-KDD datasets, the suggested model's performance is assessed [24].

The hybrid anomaly-based intrusion detection approach described by Kazem Qazanfari et al. [10] is based on SVM and Multi-Layer Perceptron (MLP). The first step in extracting features from the KDD dataset is to utilize a feature selection strategy based on the entropy of the features. The data are categorised using SVM and MLP. The outcomes of two learning-based methodologies are combined using a rule-based approach. Two-tier architecture was suggested by Divyatmika et al. to identify intrusions at the network level.

The two types of network behaviour are anomaly detection and abuse. It creates an autonomous model on the training set using hierarchical agglomerative clustering after pre-processing the data in the first step. Additionally, KNN classification is used to classify the dataset. MLP algorithm is used for misuse identification. Reinforcement algorithm is used to discover anomalies [25].

The ANN-Bayesian Net-GR approach suggested by K. Shrivastava et al. is based on ensemble and feature selection techniques. In this model, we combine the Bayesian network with the artificial neural network (ANN). Gain Ratio feature selection is used to eliminate unnecessary features and improve classification accuracy. When compared to individual models like ANN and Bayesian Net, the ensemble model is provided more accuracy. Both the KDD99 and the NSL-KDD datasets are used to assess the model [26].

A brand-new semi-supervised learning strategy based on fuzziness was put into practice by Rana Aamir et al. . Unlabeled samples are used in conjunction with a supervised learning technique to boost classifier performance. A fuzzy membership vector and class classification are produced using a Single Hidden Layer Feed-forward Neural Network (SLFNN). Each category is kept independently in the first training set by the classifier. On the basis of the NSL-KDD dataset, the experiment is assessed and contrasted with current classifier techniques.

Sumaiya T. I. et al. [14] constructed a model for multi-class classification utilizing chi-square feature selection and SVM. Radial Basis Function kernel parameter γ and over fitting constant "C" are optimised using a parameter tuning approach. The train and test times are shortened by the multi class SVM. On the basis of the NSL-KDD and KDD'99 datasets, the experiment was analysed. It showed a higher high detection rate and a lower false alarm rate when compared to other traditional approaches.

Three-layer Artificial Neural Network (ANN) model including feed-forward and backpropagation procedures, as well as different optimization strategies, was suggested by Subba et al. [15]. It retains high performance while reducing overall computational burden. The model's performance is assessed using the NSL-KDD dataset and contrasted with that of other traditional models.

For the security of the in-vehicle network, Min-Joo Kang et al. designed an effective IDS based on deep neural networks. Unsupervised deep belief networks are used to extract the probability-based feature vectors from the network packets, and then the traditional stochastic gradient descent approach is used. The DNN offers the likelihood of each class to distinguish between normal and attack packets, and as a consequence, the system is able to identify any malicious assault on the vehicle [27].

Self-taught learning was used to create a flexible and successful NIDS by Quamar Niyaz et al. . (STL). both phases A deep learning technique is STL. The first step involves learning feature representation from unlabeled data using a sparse autoencoder, and the second stage involves applying the learned representation to labelled data. The assaults on the NSL-KDD dataset are classified using the soft-max regression-based STL.

A few-shot deep learning strategy for IDS was described by Chowdhury et al. . The dataset is trained using a deep CNN structure, which is then used as a feature extractor. After that, it is trained as an SVM or 1-NN classifier to identify intrusions using the attributes that were retrieved. On the KDD and NSL-KDD datasets, the model is assessed [28].

A hybrid machine learning approach based on deep learning was suggested by Sasanka Potluri et al. [19]. Attack classes were identified using SVM and soft-max regression and complicated feature extraction was performed using a sparse autoencoder and deep belief network. In order to acquire the optimum performance rating based on the hierarchical network utilized, this model is tested on the NSL-KDD dataset [29].

The neural network ensemble technique used by Simone A. L. et al. [20] consists of autoencoder, deep belief network, deep neural network, and extreme learning machine. The assaults are classified by the autoencoder once it extracts the pertinent information. The most crucial parameters for IDSs detection rate and accuracy are used to assess the model using the NSL-KDD dataset.

The Restricted Boltzmann Machine and auto encoder were used in Zahangir et al. 's implementation of an unsupervised deep learning technique (RBM). After dimensionality reduction and feature extraction using AE and RBM, k-means clustering is used to group objects in a reduced dimension space using just three attributes. Unsupervised Extreme Learning Machine (UELML) is further used for intrusions. On the KDD'99 dataset, unsupervised methods AE and RBM with k-means clustering are tested.

In order to build a learning model and evaluate the dataset, Jin Kim et al. [22] preprocessed the data provided as input to the DNN method. Four hidden layers and 100 hidden units make up the DNN model's detection architecture. On the KDD'99 dataset, the model is put to the test [30].

In order to extract pertinent features, Ali Moradi et al. constructed a deep learning stacked autoencoder model using isolation forest as the classifier. In order to distinguish between regular and anomalous data, the SAE model pulls information from HTTP request logs.

Many variations of the LSTM architecture were suggested by Waffa Anani et al. . On an incursion dataset, tests were conducted in which several LSTM variations, including Vanilla LSTM, GRU, Bi-LSTM, and Skip-LSTM, were tested. These architectures function with the whole KDD'99 dataset. The results showed that plain LSTM performed better than alternative topologies [31].

Majjed Al-Qatf et al. deep's learning technique and self-taught learning (STL)-IDS framework were suggested in their study. For unsupervised feature reconstruction, the model is constructed using a sparse autoencoder approach. New characteristics are put into SVM after pre-training in order to increase the NSL-KDD dataset's detection and classification accuracy [32].

To increase the accuracy and efficiency, Soosan N M. et al. built a hybrid model that combines deep learning-based stacked autoencoder (SAE) with SVM. efficacy. Latent feature extraction and dataset dimensionality reduction are accomplished using SAE. SVM is a binary

Table 1

Precision score of the different classifiers on the test sets for specific devices without HDPT for DIDS.

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.76	0.75	0.77	0.78	0.77	0.67	0.81	0.78	0.72	0.73
DT	0.81	0.78	0.74	0.84	0.87	0.85	0.86	0.81	0.82	0.76
RF	0.83	0.84	0.87	0.79	0.81	0.78	0.76	0.83	0.83	0.79
KNN	0.80	0.77	0.72	0.73	0.82	0.80	0.77	0.78	0.79	0.78
GNB	0.77	0.76	0.73	0.77	0.79	0.80	0.75	0.78	0.74	0.86
TabNet	0.92	0.90	0.87	0.86	0.92	0.87	0.84	0.83	0.88	0.87
CNN	0.90	0.90	0.90	0.85	0.95	0.87	0.91	0.88	0.91	0.90

Table 2

Precision score of the different classifiers on the test sets for specific devices with HDPT for DIDS.

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.78	0.77	0.79	0.80	0.79	0.68	0.83	0.80	0.74	0.75
DT	0.82	0.79	0.75	0.86	0.89	0.87	0.88	0.82	0.84	0.77
RF	0.85	0.86	0.89	0.81	0.83	0.80	0.77	0.85	0.85	0.81
KNN	0.82	0.79	0.74	0.75	0.84	0.82	0.79	0.80	0.81	0.80
GNB	0.78	0.77	0.74	0.78	0.80	0.81	0.76	0.79	0.75	0.87
TabNet	0.93	0.91	0.88	0.87	0.93	0.88	0.85	0.84	0.89	0.88
CNN	0.92	0.92	0.92	0.87	0.97	0.89	0.93	0.90	0.93	0.92

Table 3

Recall score of the different classifiers on the test sets for specific devices without HDPT for DIDS.

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.77	0.76	0.78	0.79	0.78	0.68	0.82	0.79	0.73	0.74
DT	0.84	0.81	0.77	0.87	0.90	0.88	0.89	0.84	0.85	0.79
RF	0.85	0.86	0.89	0.81	0.83	0.80	0.78	0.85	0.85	0.81
KNN	0.79	0.76	0.71	0.72	0.81	0.79	0.76	0.77	0.78	0.77
GNB	0.78	0.77	0.74	0.78	0.80	0.81	0.76	0.79	0.75	0.87
TabNet	0.93	0.91	0.88	0.87	0.93	0.88	0.85	0.84	0.89	0.88
CNN	0.92	0.92	0.92	0.87	0.91	0.89	0.93	0.90	0.93	0.92

Table 4

Recall score of the different classifiers on the test sets for specific devices with.

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.79	0.78	0.80	0.81	0.80	0.70	0.84	0.81	0.75	0.76
DT	0.86	0.83	0.79	0.89	0.92	0.90	0.91	0.86	0.87	0.81
RF	0.87	0.88	0.91	0.83	0.85	0.82	0.80	0.87	0.87	0.83
KNN	0.80	0.77	0.72	0.73	0.82	0.80	0.77	0.78	0.79	0.78
GNB	0.80	0.79	0.76	0.80	0.82	0.83	0.78	0.81	0.77	0.89
TabNet	0.95	0.93	0.90	0.89	0.95	0.90	0.87	0.86	0.91	0.90
CNN	0.94	0.94	0.94	0.89	0.93	0.91	0.95	0.92	0.95	0.94

Table 5

F1 score of the different classifiers on the test sets for specific devices without HDPTfor DIDS.

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.71	0.70	0.72	0.73	0.72	0.62	0.76	0.73	0.67	0.68
DT	0.78	0.75	0.71	0.81	0.84	0.82	0.83	0.78	0.79	0.73
RF	0.80	0.81	0.84	0.76	0.78	0.75	0.73	0.80	0.80	0.76
KNN	0.79	0.76	0.71	0.72	0.81	0.79	0.76	0.77	0.78	0.77
GNB	0.69	0.68	0.65	0.69	0.71	0.72	0.67	0.70	0.66	0.78
TabNet	0.93	0.91	0.88	0.87	0.93	0.88	0.85	0.84	0.89	0.88
CNN	0.92	0.92	0.92	0.87	0.97	0.89	0.93	0.90	0.93	0.92

classification model that is used to categorize occurrences as either normal or abnormal. This model performs better than SVM alone in terms of accuracy and run-time when used with the ISCX IDS UNB dataset [33].

Stacked sparse autoencoder (SSAE) was used by Binghao Yan et al. to extract high-level feature representation and reduce dimensionality. The basic classifier SVM receives the low-dimensional sparse information as input. The model performed better than traditional machine learning

Table 6

F1 score sore of the different classifiers on the test sets for specific devices with HDPTfor DIDS.

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.73	0.72	0.74	0.75	0.74	0.63	0.78	0.75	0.68	0.69
DT	0.80	0.77	0.73	0.83	0.86	0.84	0.85	0.80	0.81	0.75
RF	0.81	0.82	0.85	0.77	0.79	0.76	0.74	0.81	0.81	0.77
KNN	0.81	0.78	0.73	0.74	0.83	0.81	0.78	0.79	0.80	0.79
GNB	0.70	0.69	0.66	0.70	0.72	0.73	0.68	0.71	0.67	0.79
TabNet	0.95	0.93	0.90	0.89	0.95	0.90	0.87	0.86	0.91	0.90

Table 7

Precision score of the different classifiers on the test sets for specific devices without HDPT for Attack type classification (Type of Malware).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.72	0.71	0.73	0.74	0.73	0.63	0.77	0.74	0.68	0.69
DT	0.80	0.77	0.73	0.83	0.86	0.84	0.85	0.80	0.81	0.75
RF	0.79	0.79	0.82	0.75	0.77	0.74	0.72	0.79	0.79	0.75
KNN	0.75	0.72	0.67	0.68	0.77	0.75	0.72	0.73	0.74	0.73
GNB	0.64	0.63	0.60	0.64	0.66	0.67	0.62	0.65	0.61	0.73
TabNet	0.93	0.91	0.88	0.87	0.93	0.88	0.86	0.85	0.89	0.88
CNN	0.92	0.92	0.92	0.87	0.97	0.89	0.93	0.90	0.93	0.92

Table 8

Precision score of the different classifiers on the test sets for specific devices with HDPTfor DIDS.

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.73	0.72	0.74	0.75	0.74	0.64	0.78	0.75	0.69	0.70
DT	0.82	0.79	0.75	0.85	0.88	0.86	0.87	0.82	0.83	0.77
RF	0.80	0.81	0.84	0.76	0.78	0.75	0.73	0.80	0.80	0.76
KNN	0.76	0.73	0.68	0.69	0.78	0.76	0.73	0.74	0.75	0.74
GNB	0.65	0.64	0.61	0.65	0.67	0.68	0.63	0.66	0.62	0.74
TabNet	0.95	0.93	0.90	0.89	0.95	0.90	0.87	0.86	0.91	0.90
CNN	0.94	0.94	0.94	0.89	0.99	0.91	0.95	0.92	0.95	0.94

Table 9

Recall sore of the different classifiers on the test sets for specific devices without HDPT for Attack type classification (Type of Malware).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.70	0.69	0.71	0.72	0.71	0.61	0.75	0.72	0.66	0.67
DT	0.80	0.77	0.73	0.83	0.86	0.84	0.85	0.80	0.81	0.75
RF	0.80	0.81	0.84	0.76	0.78	0.76	0.74	0.80	0.80	0.76
KNN	0.73	0.70	0.65	0.66	0.75	0.73	0.70	0.71	0.72	0.71
GNB	0.62	0.61	0.58	0.62	0.64	0.65	0.60	0.63	0.59	0.70
TabNet	0.92	0.90	0.87	0.86	0.92	0.87	0.84	0.83	0.88	0.87
CNN	0.90	0.90	0.90	0.85	0.94	0.87	0.91	0.88	0.91	0.90

Table 10

Recall score of the different classifiers on the test sets for specific devices with HDPT for Attack type classification (Type of Malware).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.71	0.70	0.72	0.73	0.72	0.62	0.76	0.73	0.67	0.68
DT	0.82	0.79	0.75	0.85	0.88	0.86	0.87	0.82	0.83	0.77
RF	0.82	0.83	0.86	0.78	0.80	0.77	0.75	0.82	0.82	0.78
KNN	0.74	0.71	0.66	0.67	0.76	0.74	0.71	0.72	0.73	0.72
GNB	0.63	0.62	0.59	0.63	0.65	0.66	0.61	0.64	0.60	0.72
TabNet	0.94	0.92	0.89	0.88	0.94	0.89	0.86	0.85	0.90	0.89
CNN	0.91	0.91	0.91	0.86	0.96	0.88	0.92	0.89	0.92	0.91

classifiers when assessed using the NSL-KDD dataset [34].

For the purpose of anomaly identification, Fahimeh. F. et al. presented the Deep Autoencoder technique. The performance of the deep model was enhanced by the model DAE using a greedy layer-wise training process. It comprises of four auto-encoders, which are sent as

input to the next layer as the current layer's output. The assaults are categorised as normal and abnormal using the softmax layer. The model was tested using the KDD'99 dataset and achieved a high detection rate [35].

To increase the rate of low-frequent attack detection, Yanqing Yang

Table 11

F1 score score of the different classifiers on the test sets for specific devices without HDPT for Attack type classification (Type of Malware).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.73	0.72	0.74	0.75	0.74	0.65	0.78	0.75	0.69	0.70
DT	0.82	0.79	0.75	0.85	0.88	0.86	0.87	0.82	0.83	0.77
RF	0.84	0.85	0.87	0.80	0.82	0.79	0.77	0.84	0.84	0.80
KNN	0.72	0.69	0.64	0.65	0.74	0.72	0.69	0.70	0.71	0.70
GNB	0.63	0.62	0.59	0.63	0.65	0.66	0.61	0.64	0.60	0.72
TabNet	0.94	0.92	0.89	0.88	0.94	0.89	0.86	0.85	0.90	0.89
CNN	0.91	0.91	0.91	0.86	0.96	0.88	0.92	0.89	0.92	0.91

Table 12

F1 score score of the different classifiers on the test sets for specific devices with HDPT for Attack type classification (Type of Malware).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.75	0.74	0.76	0.77	0.76	0.66	0.80	0.77	0.71	0.72
DT	0.84	0.81	0.77	0.87	0.90	0.88	0.89	0.84	0.85	0.79
RF	0.85	0.86	0.89	0.81	0.83	0.80	0.78	0.85	0.85	0.81
KNN	0.73	0.70	0.65	0.66	0.75	0.73	0.70	0.71	0.72	0.71
GNB	0.64	0.63	0.60	0.64	0.66	0.67	0.62	0.65	0.61	0.73
TabNet	0.96	0.94	0.91	0.90	0.96	0.91	0.88	0.87	0.92	0.91
CNN	0.92	0.92	0.92	0.87	0.97	0.89	0.93	0.90	0.93	0.92

Table 13

Precision score of the different classifiers on the test sets for specific devices without HDPT for Attack type classification (Network Disruption).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.67	0.66	0.68	0.69	0.68	0.58	0.72	0.69	0.63	0.64
DT	0.79	0.76	0.72	0.82	0.85	0.83	0.84	0.79	0.80	0.74
RF	0.79	0.80	0.83	0.75	0.77	0.74	0.72	0.79	0.79	0.75
KNN	0.74	0.71	0.66	0.67	0.76	0.74	0.71	0.72	0.73	0.72
GNB	0.65	0.64	0.61	0.65	0.67	0.68	0.63	0.66	0.62	0.74
TabNet	0.92	0.90	0.87	0.86	0.92	0.87	0.85	0.84	0.88	0.87
CNN	0.90	0.90	0.90	0.86	0.95	0.88	0.91	0.89	0.91	0.90

Table 14

Precision score of the different classifiers on the test sets for specific devices with HDPT for Attack type classification (Network Disruption).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.68	0.67	0.69	0.70	0.69	0.59	0.73	0.70	0.64	0.65
DT	0.81	0.78	0.74	0.84	0.87	0.85	0.86	0.81	0.82	0.76
RF	0.80	0.81	0.84	0.76	0.78	0.75	0.73	0.80	0.80	0.76
KNN	0.75	0.72	0.67	0.68	0.77	0.75	0.72	0.73	0.74	0.73
GNB	0.66	0.65	0.62	0.66	0.68	0.69	0.64	0.67	0.63	0.75
TabNet	0.94	0.92	0.89	0.88	0.94	0.89	0.86	0.85	0.90	0.89
CNN	0.92	0.92	0.92	0.87	0.97	0.89	0.93	0.90	0.93	0.92

Table 15

Recall score of the different classifiers on the test sets for specific devices without HDPT for Attack type classification (Network Disruption).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.68	0.67	0.69	0.70	0.69	0.59	0.73	0.70	0.64	0.65
DT	0.78	0.75	0.71	0.81	0.84	0.82	0.83	0.78	0.79	0.73
RF	0.77	0.78	0.81	0.73	0.75	0.72	0.70	0.77	0.77	0.73
KNN	0.71	0.68	0.63	0.64	0.73	0.71	0.68	0.69	0.70	0.69
GNB	0.63	0.62	0.59	0.63	0.65	0.66	0.61	0.64	0.60	0.72
TabNet	0.90	0.88	0.85	0.84	0.90	0.85	0.82	0.81	0.86	0.85
CNN	0.92	0.92	0.92	0.87	0.97	0.89	0.93	0.90	0.93	0.92

et al. adopted Supervised Variational autoencoder with regularisation Deep Neural Network (SAVAER-DNN). Instead of using a standard GAN to train the adversarial learning model, this model used WGAN-GP (Wasserstein Generative Adversarial Network with Gradient Penalty).

The encoder of SAVAER captures high-level feature representation, while the decoders create fresh attack samples [36].

To balance the train data, use the supplied label. On the NSL-KDD and UNSW-NB15 datasets, SAVAER-DNN was assessed. In order to

Table 16

Recall score of the different classifiers on the test sets for specific devices with HDPT for Attack type classification (Network Disruption).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.69	0.68	0.70	0.71	0.70	0.60	0.74	0.71	0.65	0.66
DT	0.79	0.76	0.72	0.82	0.85	0.83	0.84	0.79	0.80	0.74
RF	0.78	0.79	0.82	0.74	0.76	0.73	0.71	0.78	0.78	0.74
KNN	0.72	0.69	0.64	0.65	0.74	0.72	0.69	0.70	0.71	0.70
GNB	0.64	0.63	0.60	0.64	0.66	0.67	0.62	0.65	0.61	0.73
TabNet	0.91	0.89	0.86	0.85	0.91	0.86	0.83	0.82	0.87	0.86
CNN	0.93	0.93	0.93	0.88	0.98	0.90	0.94	0.91	0.94	0.93

Table 17

F1 score of the different classifiers on the test sets for specific devices without HDPT for Attack type classification (Network Disruption).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.65	0.65	0.66	0.67	0.66	0.57	0.70	0.67	0.62	0.63
DT	0.78	0.75	0.71	0.81	0.84	0.82	0.83	0.78	0.79	0.73
RF	0.77	0.78	0.81	0.73	0.75	0.72	0.70	0.77	0.77	0.73
KNN	0.70	0.67	0.62	0.63	0.72	0.70	0.67	0.68	0.69	0.68
GNB	0.68	0.67	0.64	0.68	0.70	0.71	0.66	0.69	0.65	0.77
TabNet	0.93	0.91	0.88	0.87	0.93	0.88	0.85	0.84	0.89	0.88
CNN	0.91	0.91	0.91	0.86	0.96	0.88	0.92	0.89	0.92	0.91

Table 18

F1 score of the different classifiers on the test sets for specific devices with HDPT for Attack type classification (Network Disruption).

	TV	Baby_Monitor	Light	Motion_Sensor	Security_camera	Smoke_Detector	Socket	Thermostat	Watch	Water_Sensor
SVM	0.67	0.66	0.68	0.69	0.68	0.58	0.72	0.69	0.63	0.64
DT	0.79	0.76	0.72	0.82	0.85	0.83	0.84	0.79	0.80	0.74
RF	0.78	0.79	0.82	0.74	0.76	0.73	0.71	0.78	0.78	0.74
KNN	0.71	0.68	0.63	0.64	0.73	0.71	0.68	0.69	0.70	0.69
GNB	0.69	0.68	0.65	0.69	0.71	0.72	0.67	0.70	0.66	0.78
TabNet	0.94	0.92	0.89	0.88	0.94	0.89	0.86	0.85	0.90	0.89
CNN	0.92	0.92	0.92	0.87	0.97	0.89	0.93	0.90	0.93	0.92

Table 19

DIDS Without applying HDPT.

	Accuracy	Precision_Score	Recall_Score	F1_Score
SVM	0.75	0.69	0.73	0.70
DT	0.81	0.78	0.78	0.78
RF	0.83	0.77	0.84	0.80
KNN	0.82	0.76	0.81	0.78
GNB	0.74	0.60	0.73	0.63
TabNet	0.93	0.91	0.93	0.90
CNN	0.89	0.90	0.92	0.89

Table 20

DIDS After applying HDPT.

	Accuracy	Precision_Score	Recall_Score	F1_Score
SVM	0.77	0.71	0.74	0.71
DT	0.82	0.79	0.79	0.79
RF	0.85	0.79	0.86	0.82
KNN	0.83	0.78	0.83	0.80
GNB	0.75	0.61	0.75	0.64
TabNet	0.95	0.93	0.95	0.92
CNN	0.91	0.92	0.93	0.91

extract the optimum feature representation and reduce the dimensionality of intrusion data, Junke Nie et al. designed the nonsymmetric Stacked Sparse Autoencoder (NSSA). The dataset was then balanced by reconstructing the output data findings. The techniques of K-means and

Table 21

Attack type classification (Type of Malware) Without applying HDPT.

	Accuracy	Precision_Score	Recall_Score	F1_Score
SVM	0.72	0.76	0.73	0.75
DT	0.83	0.81	0.81	0.81
RF	0.85	0.84	0.84	0.85
KNN	0.73	0.75	0.74	0.73
GNB	0.61	0.63	0.63	0.61
TabNet	0.94	0.93	0.91	0.93
CNN	0.92	0.92	0.91	0.92

Table 22

Attack type classification (Type of Malware) After applying HDPT.

	Accuracy	Precision_Score	Recall_Score	F1_Score
SVM	0.73	0.77	0.75	0.77
DT	0.85	0.83	0.83	0.83
RF	0.87	0.86	0.86	0.87
KNN	0.75	0.76	0.75	0.75
GNB	0.62	0.64	0.64	0.63
TabNet	0.96	0.95	0.92	0.95
CNN	0.94	0.94	0.93	0.94

Mean Shift were coupled for feature extraction. The assaults are categorised using the random Forest classifier. In order to shorten the train time, the model balances the ISCX data and extracts the features [37].

The convolutional neural network suggested by Guojie Liu et al. and

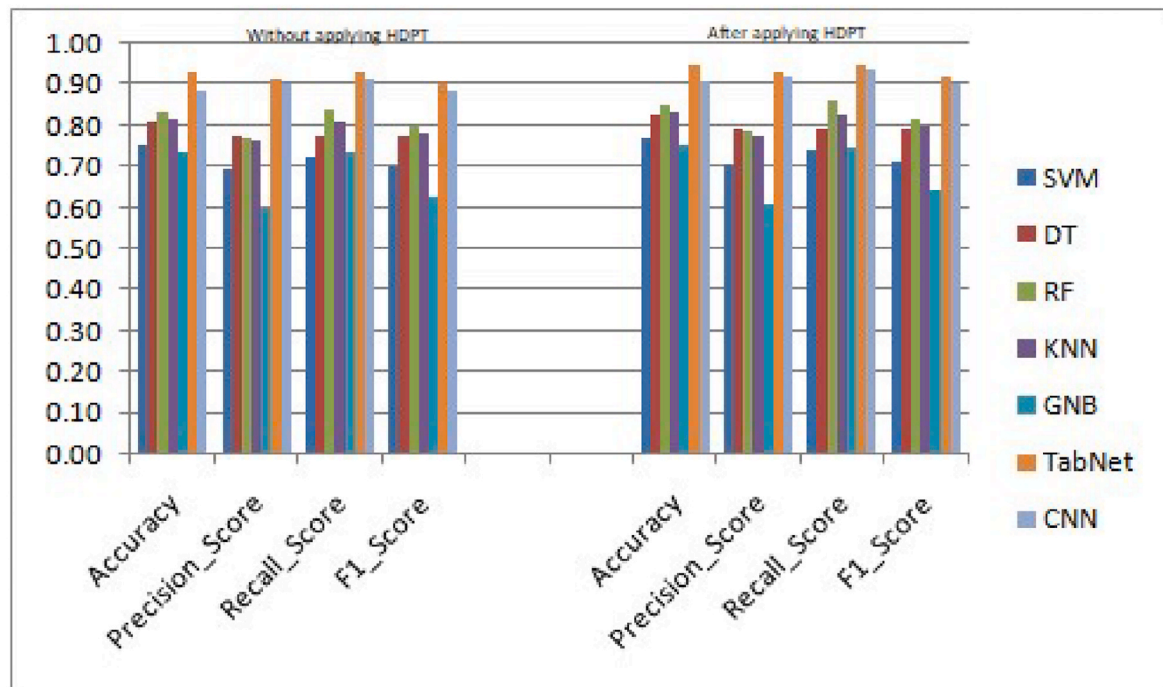


Fig. 2. Overall comparison of different performance parameters for DIDS.

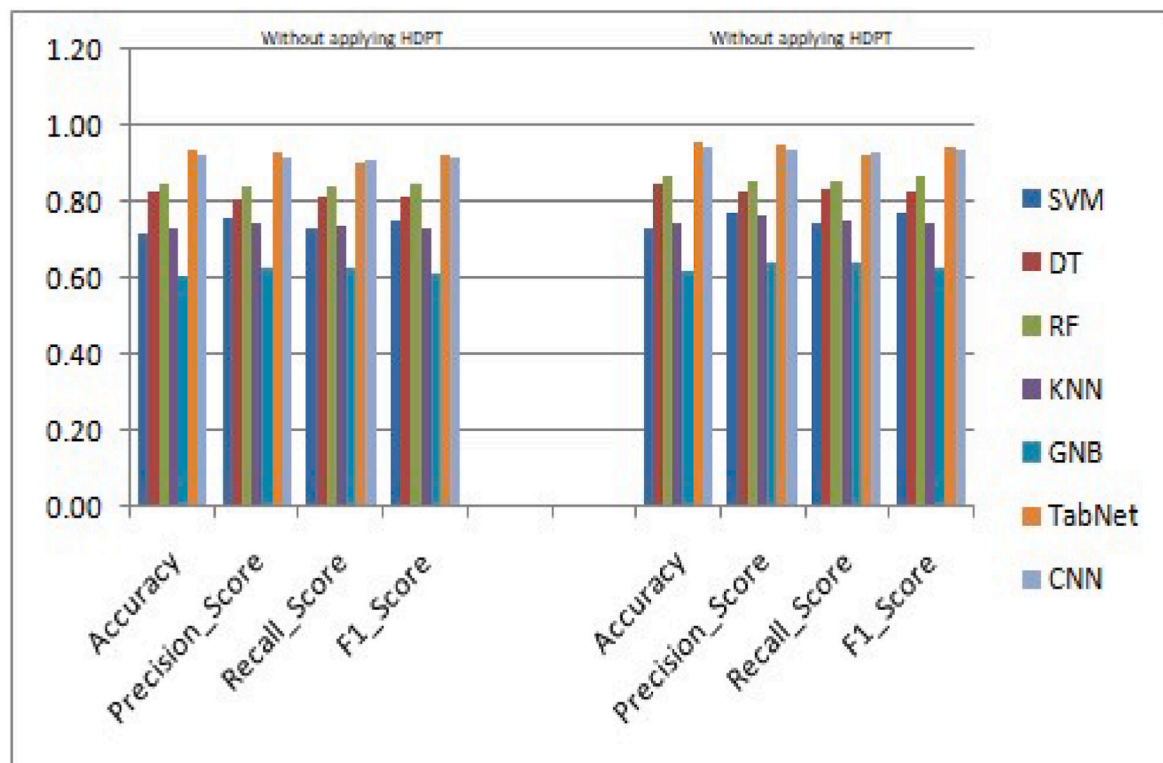


Fig. 3. Comparison of different performance parameters for Attack type classification (Type of Malware).

the method are optimised. The original one-dimensional network intrusion data is pre-processed and transformed into two-dimensional data, after which the CNN and softmax classifiers use the model's self-learned effective features to categorize the multiclass assaults. The mode was assessed using the NSL-KDD and KDD-CUP 99 datasets [38].

To improve the model's accuracy, Wen-Hui Lin et al. introduced an IDS model based on a CNNs-based classifier. In order to optimise the

model parameters and reduce classification errors, this method revises the LaNet-5 model using the adaptive delta optimization technique. The model's accuracy was enhanced while being assessed on the KDD'99 dataset [39].

For the purpose of detecting intrusions, Naila Marir et al. suggested a brand-new semi-supervised distributed technique based on stacked denoising sparse autoencoder and SVM. This method uses a stacked de-

noising autoencoder to carry out a non-linear unsupervised dimensional reduction. The SVM is used to categorize invasions. The iterative reduction methodology built on Apache Spark is used to implement the model. The experiment assessed the promisingly performing KDD Cup'99, NSL-KDD, UNSW-NB15, and CICIDS2017 datasets [40].

In order to create effective IDSs, M. Kasongo et al. investigated the use of the XGBoost strategy for feature selection and integrated it with many machine learning techniques, including ANN, k-nearest neighbour, Decision Tree, Linear Regression, and SVM. The models took into account binary and multiple attack categorization while evaluating the UNSW-NB15 dataset. The results showed that the accuracy was boosted by the features chosen using the XGBoost-based approach and decision tree classification model [41].

In order to choose the most appropriate features and utilize CNN for classification, Riyaz et al. suggested IDS to include a new feature selection technique called conditional random field and linear correlation coefficient-based feature selection (CRF-LCFS) algorithm. Additionally, CNN is beneficial for enhancing performance with the KDD'99 dataset.

3. Proposed methodology

3.1. IOT network device based intrusion detection system

The growing market for Internet of Things (IoT) devices creates new challenges for tracking computer network activity. The tremendous variety of smart devices cannot be accommodated by traditional network monitoring. New network discovery techniques must be created to identify the Internet of Things (IoT) devices that are connected to the network. In this situation, the use of data analysis tools can be made to look for specific patterns that help identify various device kinds. In actuality, unlike desktop computers, Internet of Things devices perform fairly specific tasks, which makes the networking behaviour of these devices quite predictable. This paper proposes a machine learning-based method for classifying the different Internet of Things device types connected to a network by examining streams of packets sent and received on that network.

To gather information regarding network traffic, a trial smart home network was setup, a model to characterize the behaviors of IoT device networks using the data that was gathered was developed. By visualizing the data with the help of the t-SNE technique, the network traffic can be distinguished that is caused by the various Internet of Things devices. Following this, five different machine learning classifiers are trained using data characterizing the network activities. and two Deep learning methods to predict the Internet of Things device that was responsible for the network traffic.

The research on the current state of the art revealed some of the shortcomings of the machine learning-based Internet of Things network monitoring technologies that are now in use. Attempts to identify devices connected to the Internet of Things have not been successful in being phase independent, delay-free, or nonintrusive there the Device based Intrusion Detection System's efforts are limited, this could be because it could only detect certain kinds of attacks or it requires knowing the type of device which generates network traffic data for functioning properly (while it could have been a completely different scenario if in case it was deployed outside the local network).

In this objective, the main target is to develop a machine learning-based Internet of Things network tool for monitoring purposes and to overcome the limitations of existing works. The investigation is carried out in the following two categories as discussed below:

3.1.1. IoT's device type recognition system

The goal of the system is to identify the type of device by analyzing the network traffic that the device generates on the network. The system that is being created needs to have no delay, phase independence, and be non-intrusive. To begin, the process of describing bidirectional flows begins with the definition of a collection of characteristics that adhere to

the aforementioned limitations. After that, A smart home to conduct experiments, which consists of various smart IoT devices was established for generating real-time data on the traffic of an Internet of Things network. The purpose of performing data visualization with t-sne understand the data which has bidirectional flows and to evaluate how efficient the chosen set of characteristics is in distinguishing between the various IoT devices. The next step is to train a variety of machine learning-based algorithms for classifying the bidirectional flows based on the type of Internet of things to which they belong.

3.1.2. IoT device based intrusion detection system

Its goal is to identify abnormal behaviour, the system should have the capability to identify different kinds of attacks, without letting it know the type of connected device which is generating network traffic while performing its functionalities, because there is such a wide variety of IoT devices (cameras, smart bulbs, motion sensors, and so on), how they interact with network communications might vary substantially from one item to the next. As a result, one unique learning is done for each different kind of Internet of Things device.

During the phase of testing, investigation of two different possibilities one is identifying the type of device connect the other is identifying the abnormal behaviour in data based on the network traffic data and the other identifies the intrusion.

3.1.3. Device type recognition system

To improve the effectiveness of the network security measures, it will be very much helpful to identify the type of device connected to the network [2]. For instance, The proper filtering rules can be established once it has been determined that the device in question is a security camera produced by a particular manufacturer. This ensures that the camera will not be able to carry out any actions that are not in accordance with what is expected of it and will be restricted to only those actions that are in line with what is expected of it. In addition, the identification of the type of device that is trying to connect to the network can be used to block access to the network for blacklisted devices.

The recognition of different types of IoT devices can also be used for harmful purposes. For instance, an attacker could find susceptible Internet of Things devices by performing passive network traffic analysis. Because IoT networks are wireless, an adversary can easily intercept the traffic on the network to carry out additional analysis with the intention of the process of determining the types of devices that are currently connected to the network [35]. The passive detection of vulnerable devices by malware can also be assisted by device fingerprinting accomplished through the monitoring of network data. Passively locating vulnerable devices, as opposed to seeking a device to actively infect, will minimize the network signature of the malware, making it even more difficult for intrusion detection systems to spot it. The recognition of the type of device also presents privacy problems [36]. An intruder may be able to discover the present status of a device after first identifying the type of device it is by conducting further analysis of the traffic and using that information. For instance, in, multi-class classification is carried out so that network traces of a device can be associated with the appropriate condition of the device.

Because there are such a wide variety of devices connected to the internet of things, It is difficult to come up with a one-of-a-kind network signature for every category of goods. Consequently, algorithms that use machine learning are better prepared to learn patterns in the network traffic and differentiate [2] between the various types of devices. This is because machine learning algorithms can more easily learn large amounts of data. The current state of the art has brought to light the limits of previous efforts on the classification of Internet of Things network data using machine learning. Because certain works take features from the entirety of the transmission without making use of any timeouts, the user needs to wait for a significant period before they can determine the type of device that is being used. Other initiatives have

restrictions as a result of the fact that they focus on a particular stage of the life cycle of a device or a specific kind of network packet. This limits the scope of the problem that can be solved. For instance, the setup phase of an Internet of Things device is where the type of device is identified [36]. Because of this, it is not possible to apply the proposed method if failed to complete the setup phase of the process, to identify the type of device which are already connected to the network. Several works are restricted in some way because they make use of application-level features that have been extracted from the system. This necessitates that the network communication is unencrypted, which is another limitation. Privacy concerns are also raised when looking at the information at the application level.

An IoT device type recognition system by conducting a network traffic analysis, for overcoming the existing model limitations is proposed the method is shown in Fig. 1.

To begin, a feature set is determined that adheres to the constraints that were previously outlined. Through the use of visualization tools, whether or not the set of attributes that has chosen is effective enough to discern between the various sorts of IoT devices can be determined. A device model produced by a certain manufacturer when using the term "IoT device type." After that, network traffic classifiers that are based on machine learning are taught to determine the type of IoT devices that are connected to a network. The workflow of the newly built model is depicted in Fig. 1. Before relevant features can be extracted from raw network traffic, they must first be preprocessed. After the features have been retrieved, they are inputted into a classifier to determine the kind of Internet of Things device that was responsible for generating the network traffic.

By 2030, the overall number of IoT devices will drastically be increased to 75 billion as per an estimate. Network administrators face additional hurdles as the Internet of Things business expands. With such a varied spectrum of IoT devices, device recognition is crucial.

The development of IoT threats such as Mirai has made network security for IoT devices vital. Mirai infected around 600,000 devices worldwide at its height. DDoS attacks were launched by hackers using hacked equipment. Here knowing what type of devices is connected, will aid in security implementation, knowing that a device is of a certain brand, for example, can assist a network administrator in creating filtering rules to prohibit the camera from doing anything other than what it was intended to do. Identification of device type can also be used to prevent potentially vulnerable devices from joining the network.

Through passive network traffic analysis, an attacker could use IoT device recognition to locate vulnerable IoT devices. Even if the connection is encrypted, an attacker can easily access network data from a wireless IoT network and conduct additional analysis to determine the type of connected devices.

Malware can also detect vulnerable PCs invisibly using network traffic analysis and device fingerprinting, rather than seeking permissions from vulnerable devices to infect. Passive devices used for vulnerability detection decrease the malware's network signature, making intrusion detection systems even more difficult. The recognition of device type presents privacy problems. After identifying a device, the methods mentioned in can be used to establish its current status. This type of information might be used to deduce what is going on in a smart home, potentially revealing personal information.

A gadget that is linked to the Internet of Things fulfills a specific function. IoT devices, unlike laptops, desktop computers, or smart phones, execute incredibly particular jobs. A smart bulb or a smart switch, for example, can be turned off or on. A smart light's brightness can also be modified. In contrast, a smart lamp or socket is not designed to play YouTube videos nor it can send an email. As a result, network traffic from IoT devices has a unique pattern that is predictable; machine learning techniques perform well with it.

A method for recognizing IoT devices is presented using network data in this article. Bidirectional flows obtained from raw network data are described by the N packet size broadcast and received. Due to a lack

of publicly available data, The data of smart homes is taken for conducting experiments and analyses for identifying types of IoT devices. t-SNE is then used to visualize and analyze the data. Furthermore, the visual clusters obtained demonstrate that the set of criteria, distinguish between distinct IoT devices is effective. Then, to categorize network traffic, different DL and ML algorithms are implemented and evaluated.

As a result of the increasing proliferation of IoT networks, a small number of studies related to IoT device fingerprinting have emerged. Network traffic analysis based on machine learning technique for recognizing IoT devices and generating authorized devices list. They work with data from entire TCP sessions T. D. Nguyen and colleagues developed a method for detecting hacked Internet of Things devices. They take advantage of the generated network traffic's temporal periodicity. The devices can be identified using periodic flow characteristics, duration, accuracy, and stability. To start with profiles for standard communication for specific devices are created. An RNN (Recurrent Neural Network) is used to detect deviations in predicted behaviour. Describe a method for recognizing the IoT device type connected to a network and preventing vulnerable devices from connecting. They take advantage of several features gathered during device configuration. In contrast to previous systems, B. Bezawada and colleagues proposed a method for fingerprinting device behaviour. As a feature is used I form of a subset as well as payload-based attributes are used to train the model. R. Doshi et al. detect DDOS attacks in IoT networks using network traffic classification.

The work on IoT device identification is notable for employing a novel collection of attributes that can be recovered even from encrypted network traffic. Other techniques, such as obtaining sensitive data about the current status of a device, address privacy issues. As a result, A. Acar et al. show how an attacker can utilize machine learning methods to profile network data to determine current user behaviour in a smart home. Analyzing a network's traffic flow rates can reveal user behaviour which is very sensitive. An attacker may use the transmission and reception rates of the CCTV camera to detect movement in a smart home. look at a few of the popular devices, the Thermostat and a wired Nest Protect, to understand how an attacker can easily monitor network traffic to determine whether a home is occupied. Rather than focusing on privacy issues, the primary goal is to create a system for recognizing the IoT devices which are connected to the network.

4. Result section

Internet of Things is one of the fastest-growing fields which provides the scope of interconnectivity, IoT devices can be interconnected with echo the and can also transfer information from one place to another because of their unique properties but IoT devices have lacks important features of security to deal with the cyber attacks as these devices do not have dedicated memory space it is not possible to implement anti-malware or any ant viruses but by using machine learning this problem can be tackled, Machine Learning algorithms if trained have very have the unique quality of identifying and predicting the attacks. In this thesis similar kinds of problems are explored and trying to resolve them with the help of Machine Learning, to do this the dataset will be used of a smart home in which is been analyze on various smart devices like a smart bulb, door bell, CCTV camera, temperature sensors, wifi routers, etc. The result analysis is shown in Table 1 - 22 and Fig. 2, Fig. 3.

4.1. Results for monitoring (DIDS)

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4.2. Results for attack type classification (type of malware)

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4.3. Results for attack type classification (network disruption)

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4.4. Results for DIDS

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5. Results for attack type classification (type of malware)

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6. Conclusion & future scope

When it comes to DIDS, methods like Device Type Identification and Device-based Intrusion Detection System are employed, along with various machine learning and Deep Learning methods. Their performance is evaluated, and it is discovered that Random forest is the best performing algorithm for DIDS with 95% accuracy and Random forest is best performing for identifying the type of device with 3% accuracy. An whole new level of network security is given by attack type categorization. The classification and identification of unknown network disturbances and attack kinds (malware type) using machine learning and deep learning approaches is the focus of this thesis. With accuracy of 96% for recognizing the kind of malware and 95% for identifying network disturbances, TabNet was determined to be the overall top performing algorithm in this method, wh When it comes to DIDS, techniques such as Device Type Identification and Device based Intrusion Detection System are used, as well as different machine learning and Deep Learning methods are applied and their performance are evaluated and found the Random forest is best performing for identifying the type of device with 3% accuracy and Tabnet is best performing algorithm for DIDS with 95% accuracy.

In attack type classification, a new realm of network security is delivered. Machine learning and Deep Learning techniques to classify and identify unknown network disruptions and types of attacks (malware type) in this thesis. This strategy fully utilizes the benefits of Machine Learning as well as Deep Learning algorithms and TabNet was found to be overall best performing algorithm with accuracy of 96% for identifying type of malware and accuracy of 95% for identifying the disruptions in the network. ich fully exploits the advantages of both machine learning and deep learning techniques.

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The author 's declare no contribution.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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